EE 2001.1 Experiment 3

Group no: 11.1

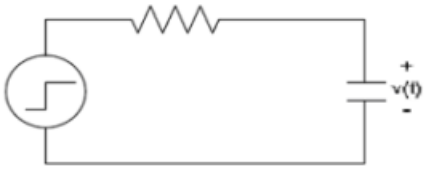
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construct the circuit, adjust assiloscope

Group member 2: Mehmet Ali Kamalak, 150720062, Mehmet
calculating, construct the circuit

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Calculating output and input

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function generators, adjust osscilOSLope

Q1) TO DO PHYSICALLY AT THE LAB (Use: R = 3.3 kΩ, C = 22nF) Build a first order RC circuit as below, adjust voltage square wave with peak value of 5 V at 1 kHz, observe and record input and output one on the other using oscilloscope.



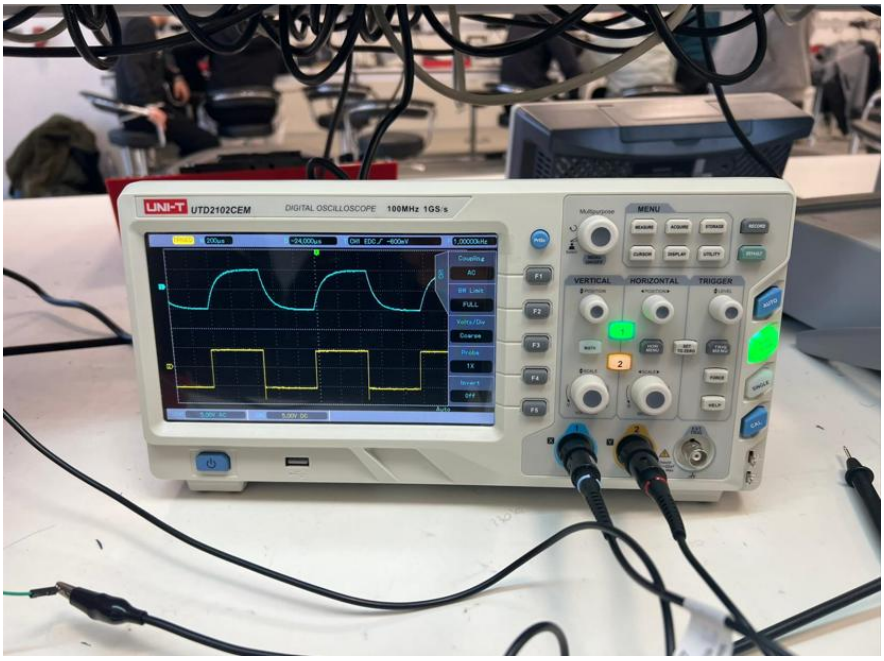


Figure 1: input and output(R=3.3k)

Q2)

The time constant (τ) of the circuit was initially calculated using the formula τ=RC, where R is the resistance and C is the capacitance. In our case, τ was determined as

τ=(3.3kΩ)(22nF)=72.6µs

Alternatively, measuring the time to peak voltage yielded an approximate value of 42µs for τ. The discrepancy between the two values may be due to variations in circuit components or measurement inaccuracies, such as the oscilloscope reading 4.40V and 0.89kHz instead of the intended values

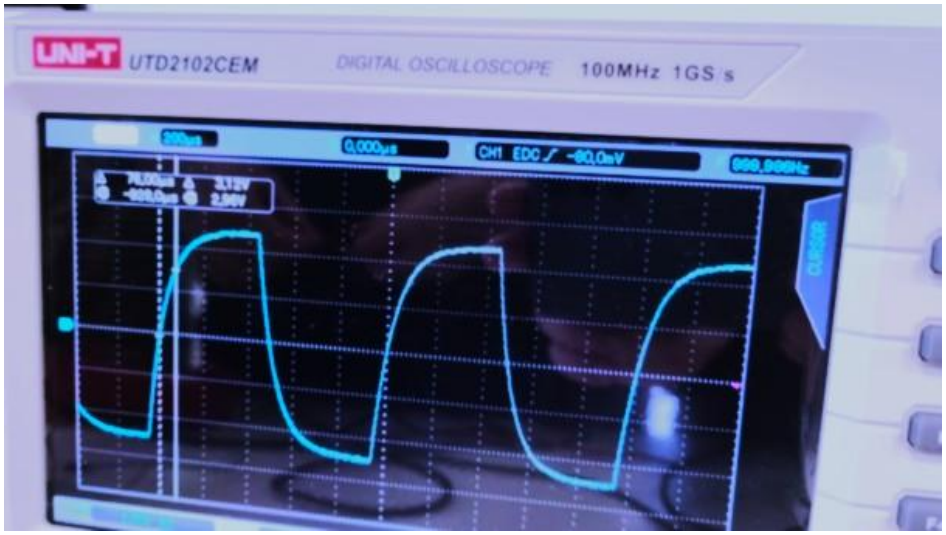


Figure 2 :Time Constant for 3.3k

Q3)

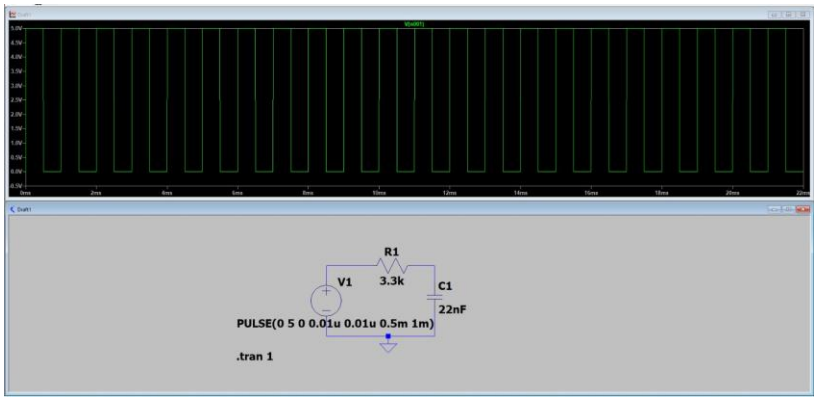


Fig. 1: The Circuit 1

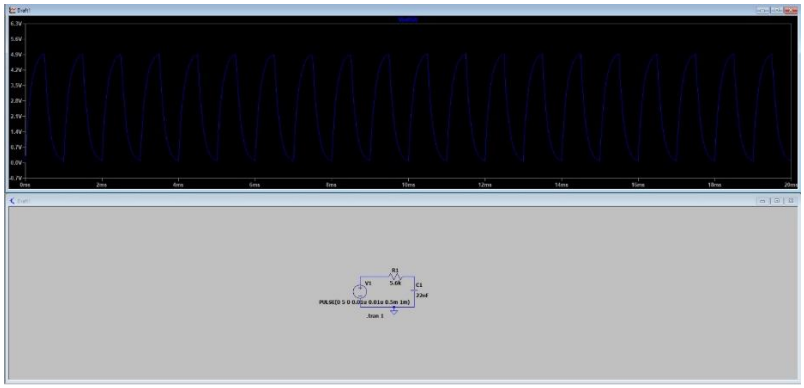
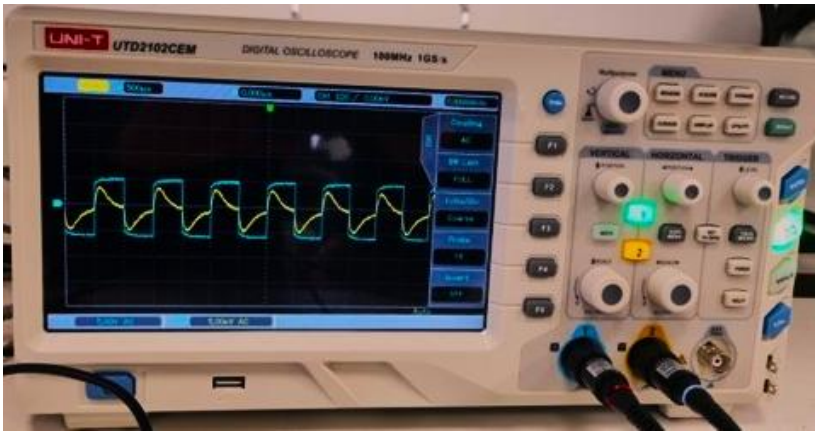
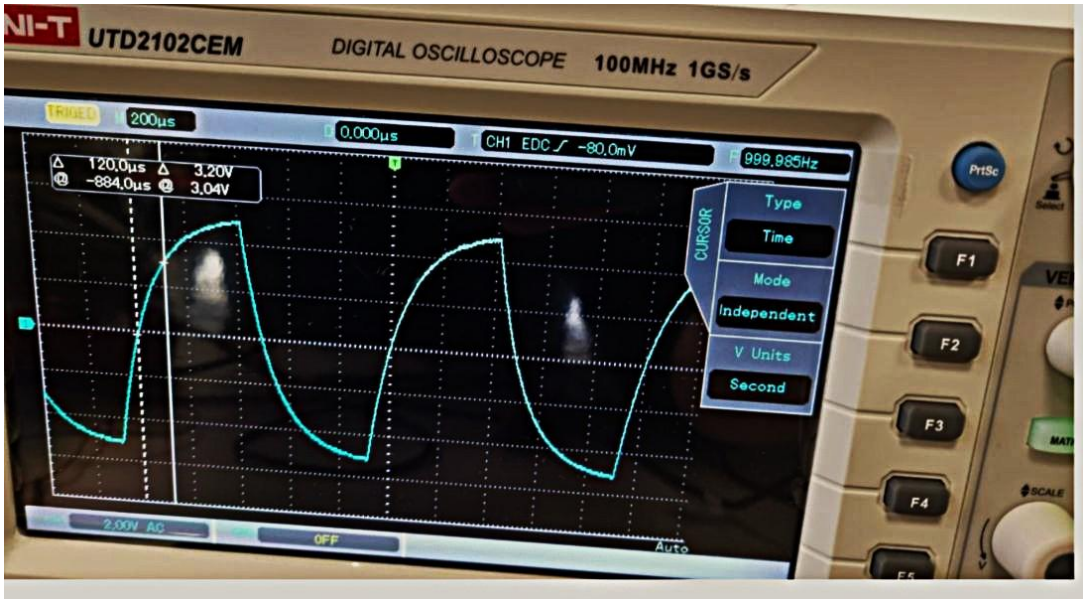


Fig. 2: The Circuit 2



5.6 k Circuit Output



Time Constant for 5.6k

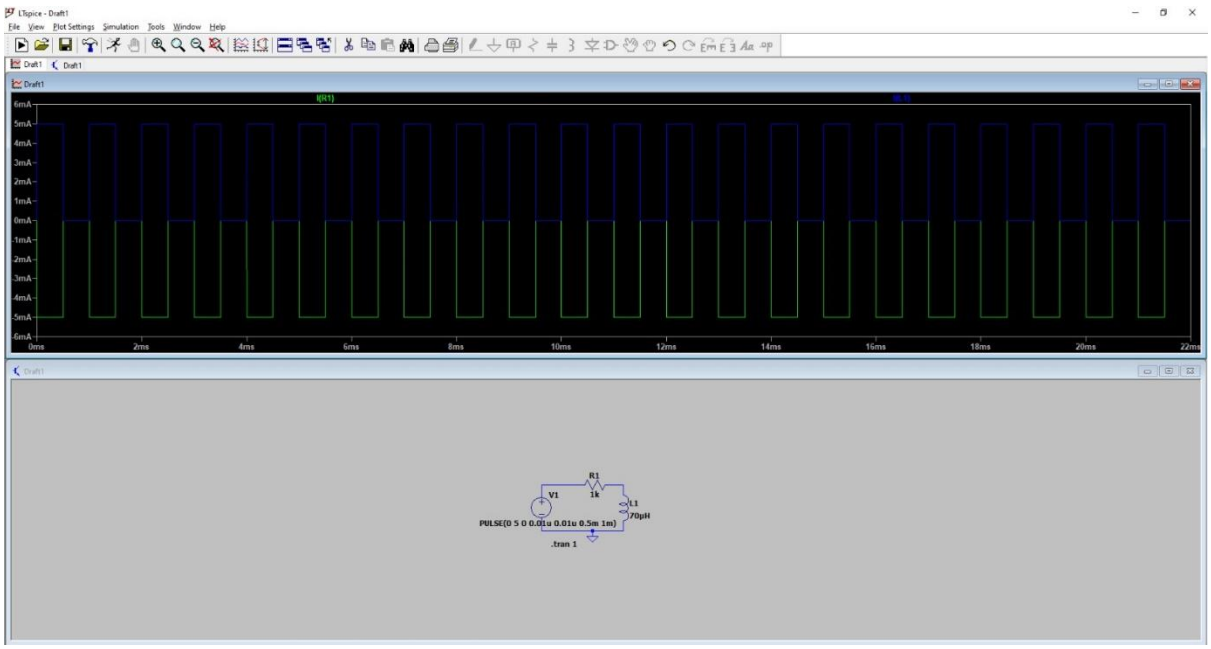
5.6k Time Constant : 1232*10(-7)

3.3k Time Constant : 726*10(-7)

5.6k Rise Time=270us Fall time= 210us

3.3k Rise Time=170us Fall Time=150us

Q4)



Q5) Time constant explanation is τ=L/R ;

τ= (70 µH) / (1Ω) = 10 µs

Current explanation is i=V/R ;

i = (5V)/(1Ω)=5A

i(t)=i (1-e^((-5τ/τ)))

i(t)=5A(1-e^((-5τ/τ))) = 4.96631A

Q6)

